

Spatial Deconfounder

Interference-Aware Deconfounding for Spatial Causal Inference



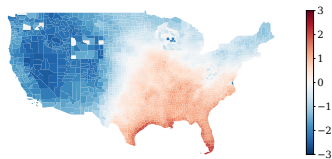
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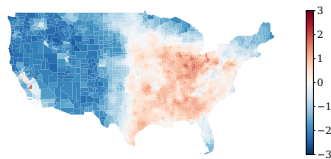
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Spatial Causal Inference

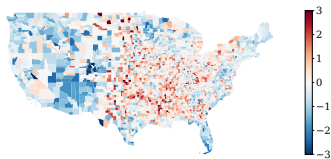
- Spatial index $s = (i, j)$
- Neighborhood $\mathcal{N}_s$ is a $(2r + 1) \times (2r + 1)$ square centered at s , excluding s itself
- **Features (Covariates)** $\mathbf{X}_s, \mathbf{X}_{\mathcal{N}_s}$
- **Unobserved confounder** $U(s)$
- **Interventions (Treatments)** $A_s, A_{\mathcal{N}_s} \in \{0, 1\}$
- **Outcomes** Y_s



Humidity ($U(s)$)



$PM_{2.5}$ (A_s)



Mortality (Y_s)

Two Intertwined Challenges

1 Interference (SUTVA violation)

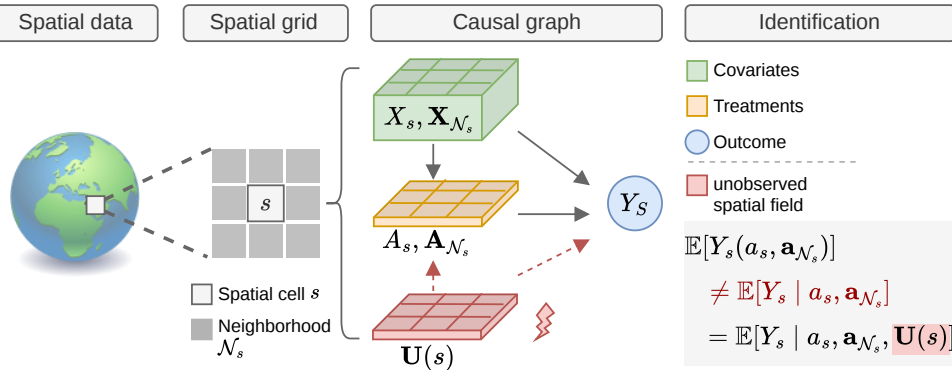
$$Y_s(\mathbf{a}) = Y_s(a_s, \mathbf{a}_{\mathcal{N}_s})$$

Pollution / disease spill across regions

2 Unobserved spatial confounding

$U(s)$ (humidity, topography, ...)

Jointly drives A_s and Y_s

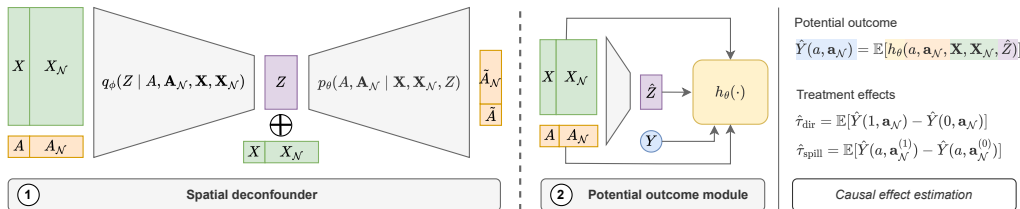


Existing methods handle one challenge by *assuming away* the other.

We show that they are deeply connected:
interference reveals structure in the latent confounder.

Each unit observes $(A_s, \mathbf{A}_{\mathcal{N}_s})$, all shaped by the *same* $U(s)$

- The deconfounder methods (Wang & Blei, 2019) assume i.i.d units with multiple simultaneous causes.
- Localized interference creates the multi-cause structure needed for deconfounding
- This yields an interference-driven deconfounding strategy that reconstructs latent spatial confounding *nonparametrically* and *without multiple treatment types*

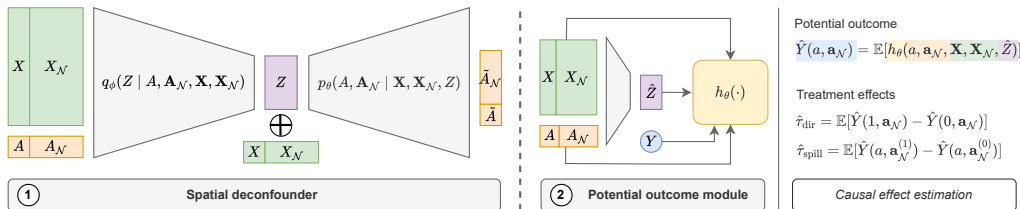


Stage 1: Confounder Reconstruction with C-VAE

- **Encoder** maps treatments and covariates into latent embedding Z_s :

$$q_\phi(Z_s | A_s, \mathbf{A}_{N_s}, X_s, \mathbf{X}_{N_s}) = \mathcal{N}(\mu_\phi(\cdot), \text{diag } \sigma_\phi^2(\cdot))$$

- **Decoder** predicts A_s given covariates and latent: $p_\psi(A_s | X_s, \mathbf{X}_{N_s}, Z_s)$
- **GMRF prior** enforces spatial smoothness: $p_\theta(Z) = \mathcal{N}(\mathbf{0}, \tau^{-1}(L + \epsilon I)^{-1})$



Stage 2: Potential outcome module

- After Stage 1, $\hat{Z}_s = \mathbb{E}_{q_{\phi}}[Z_s]$ and using a **flexible spatial model** h (e.g. U-Net):

$$\hat{Y}_s = h(A_s, \mathbf{A}_{\mathcal{N}_s}, X_s, \mathbf{X}_{\mathcal{N}_s}, \hat{Z}_s)$$

- Direct Effect:** $\hat{\tau}_{\text{dir}} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} [h(1, A_{\mathcal{N}_s}, \mathbf{X}_s, \mathbf{X}_{\mathcal{N}_s}, \hat{Z}_s) - h(0, A_{\mathcal{N}_s}, \mathbf{X}_s, \mathbf{X}_{\mathcal{N}_s}, \hat{Z}_s)]$
- and analogously for the spillover effect $\hat{\tau}_{\text{spill}}$

Under **localized interference** and typical **deconfounding assumptions** (e.g. no single-cause confounding), we can reconstruct a confounder Z_s such that the direct and spillover effects are identifiable as

$$\tau_{\text{dir}} = \mathbb{E}_{\mathbf{X}_s, \mathbf{X}_{\mathcal{N}_s}, Z} \left[\mathbb{E}_Y[Y_s \mid A_s = 1, \mathbf{A}_{\mathcal{N}_s}, \mathbf{X}_s, \mathbf{X}_{\mathcal{N}_s}, Z_s] - \mathbb{E}_Y[Y_s \mid A_s = 0, \mathbf{A}_{\mathcal{N}_s}, \mathbf{X}_s, \mathbf{X}_{\mathcal{N}_s}, Z_s] \right], \quad (1)$$

$$\tau_{\text{spill}} = \mathbb{E}_{\mathbf{X}_s, \mathbf{X}_{\mathcal{N}_s}, Z} \left[\mathbb{E}_Y[Y_s \mid a, \mathbf{A}_{\mathcal{N}_s} = \mathbf{a}_{\mathcal{N}_s}^{(1)}, \mathbf{X}_s, \mathbf{X}_{\mathcal{N}_s}, Z_s] - \mathbb{E}_Y[Y_s \mid a, \mathbf{A}_{\mathcal{N}_s} = \mathbf{a}_{\mathcal{N}_s}^{(0)}, \mathbf{X}_s, \mathbf{X}_{\mathcal{N}_s}, Z_s] \right]. \quad (2)$$

SpaCE (Tec et al., 2024): semi-synthetic spatial causal datasets from real datasets

Our extensions:

- Generate outcomes under two confounding regimes:

$$\text{(Local confounding)} \quad \hat{Y}_s = f(A_s, A_{\mathcal{N}_s}, X_s) + R_s, \quad (3)$$

$$\text{(Spatial confounding)} \quad \hat{Y}_s = f(A_s, A_{\mathcal{N}_s}, X_s, X_{\mathcal{N}_s}) + R_s, \quad (4)$$

where f is a predictive function learned from the observed data

- Mask key covariates to simulate hidden $U(s)$

Datasets: $PM_{2.5} \rightarrow$ Mortality, (socioeconomic) $SO_4 \rightarrow PM_{2.5}$ (atmospheric)

Baselines:

- Spatial autoregressive models (S2SLS-LAG1, DURBIN, GMERROR)
- Spline-based SPATIAL and SPATIAL+ with exposure-mapped variants *E-MAP*
- Other models like GCNN, DAPSM, and UNET

Results: Local Confounding

Table 1: Standardized absolute bias, $\sigma_y^{-1} |\hat{\tau} - \tau| \pm 95\%$ CI. $p \approx 0.5$: good model fit; R : deconfounder neighborhood radius. Best and second-best values are bolded and underlined, respectively. **Environment:** $PM_{2.5} \rightarrow m$ ($r_d = 1$).

hidden confounder: ρ_{pop}

Method	DIR	SPILL	p
C-VAE-SPATIAL+ ($R = 1$)	0.02 $\pm$ 0.01	0.02 $\pm$ 0.00	0.51 $\pm$ 0.07
C-VAE-SPATIAL+ ($R = 2$)	0.04 $\pm$ 0.01	<u>0.04 $\pm$ 0.01</u>	0.50 $\pm$ 0.05
DAPSM	0.25 $\pm$ 0.01	n/a	n/a
GCNN	0.36 $\pm$ 0.03	n/a	n/a
GMERROR	<u>0.03 $\pm$ 0.00</u>	n/a	n/a
DURBIN	<u>0.03 $\pm$ 0.00</u>	0.07 $\pm$ 0.00	n/a
S2SLS-LAG1	<u>0.03 $\pm$ 0.00</u>	0.07 $\pm$ 0.00	n/a
SPATIAL+	0.05 $\pm$ 0.02	n/a	n/a
SPATIAL	0.02 $\pm$ 0.00	n/a	n/a
E-MAP-SPATIAL+ ($R = 1$)	0.04 $\pm$ 0.02	0.07 $\pm$ 0.07	n/a
E-MAP-SPATIAL ($R = 1$)	0.02 $\pm$ 0.00	0.07 $\pm$ 0.07	n/a

hidden confounder: q_{summer}

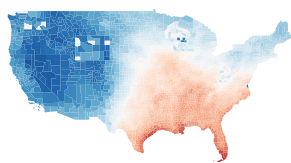
Method	DIR	SPILL	p
C-VAE-SPATIAL+ ($R = 1$)	0.02 $\pm$ 0.01	0.02 $\pm$ 0.00	0.53 $\pm$ 0.05
C-VAE-SPATIAL+ ($R = 2$)	<u>0.04 $\pm$ 0.01</u>	<u>0.04 $\pm$ 0.01</u>	0.51 $\pm$ 0.05
DAPSM	0.30 $\pm$ 0.03	n/a	n/a
GCNN	0.41 $\pm$ 0.03	n/a	n/a
GMERROR	0.20 $\pm$ 0.00	n/a	n/a
DURBIN	0.20 $\pm$ 0.00	0.06 $\pm$ 0.00	n/a
S2SLS-LAG1	0.20 $\pm$ 0.00	0.06 $\pm$ 0.00	n/a
SPATIAL+	0.05 $\pm$ 0.02	n/a	n/a
SPATIAL	0.02 $\pm$ 0.00	n/a	n/a
E-MAP-SPATIAL+ ($R = 1$)	0.05 $\pm$ 0.02	0.07 $\pm$ 0.07	n/a
E-MAP-SPATIAL ($R = 1$)	0.02 $\pm$ 0.00	0.07 $\pm$ 0.07	n/a

Results: Spatial Confounding

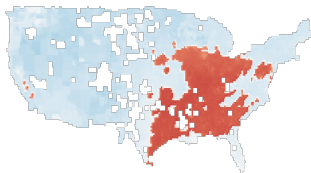
Table 2: Standardized absolute bias, $\sigma_y^{-1} |\hat{\tau} - \tau| \pm 95\% \text{ CI}$. $p \approx 0.5$: good model fit; R : deconfounder neighborhood radius. Best and second-best values are bolded and underlined, respectively.

$PM_{2.5} \rightarrow m, (r_d = 1), \text{ conf: } \rho_{\text{pop}}$				$SO_4 \rightarrow PM_{2.5}, (r_d = 1), \text{ conf: } OC$			
Method	DIR	SPILL	p	Method	DIR	SPILL	p
C-VAE-UNET ($R = 1$)	0.06 ± 0.02	0.16 ± 0.08	0.53 ± 0.03	C-VAE-UNET ($R = 1$)	0.09 ± 0.00	0.12 ± 0.02	0.48 ± 0.05
C-VAE-UNET ($R = 2$)	<u>0.04 ± 0.01</u>	0.07 ± 0.02	0.51 ± 0.04	C-VAE-UNET ($R = 2$)	0.06 ± 0.01	0.07 ± 0.03	0.54 ± 0.03
DAPSM	0.20 ± 0.01	n/a	n/a	DAPSM	1.57 ± 0.00	n/a	n/a
GCNN	0.17 ± 0.06	n/a	n/a	GCNN	0.42 ± 0.15	n/a	n/a
GMERROR	0.05 ± 0.00	n/a	n/a	GMERROR	0.13 ± 0.00	n/a	n/a
DURBIN	0.05 ± 0.00	<u>0.05 ± 0.00</u>	n/a	DURBIN	0.13 ± 0.00	0.01 ± 0.00	n/a
S2SLS-LAG1	0.05 ± 0.00	<u>0.04 ± 0.00</u>	n/a	S2SLS-LAG1	0.13 ± 0.00	0.08 ± 0.00	n/a
SPATIAL+	0.12 ± 0.09	n/a	n/a	SPATIAL+	0.12 ± 0.08	n/a	n/a
SPATIAL	0.03 ± 0.00	n/a	n/a	SPATIAL	<u>0.07 ± 0.00</u>	n/a	n/a
UNET	0.06 ± 0.01	0.17 ± 0.04	n/a	UNET	<u>0.07 ± 0.02</u>	0.05 ± 0.02	n/a
E-MAP-SPATIAL+ ($R = 1$)	0.11 ± 0.09	0.06 ± 0.06	n/a	E-MAP-SPATIAL+ ($R = 1$)	<u>0.10 ± 0.07</u>	<u>0.05 ± 0.01</u>	n/a
E-MAP-SPATIAL ($R = 1$)	0.03 ± 0.00	0.07 ± 0.06	n/a	E-MAP-SPATIAL ($R = 1$)	0.07 ± 0.00	<u>0.05 ± 0.01</u>	n/a

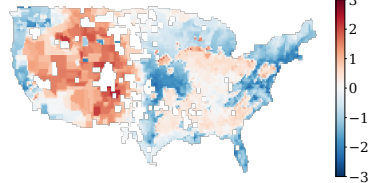
Recovering the Hidden Spatial Field



True $U(s)$
(summer humidity q_{summer})



PC1 of $\hat{Z}_s$
(captures treatment)



PC2 of $\hat{Z}_s$
(slightly recovers $U(s)$)

The second principal component of the latent space mirrors the large-scale spatial structure of the true (unobserved) confounder

Key Contributions:

- **Spatial Deconfounder:** A framework that **jointly addresses interference and unobserved spatial confounding** by treating neighborhood treatments as a multi-cause signal.
- We establish **identification from a C-VAE** with a spatial prior under assumptions on the **latent spatial field** and **outcome structure**.
- **Empirics:** In semi-synthetic datasets with both interference and unobserved spatial confounding, the **Spatial Deconfounder shows consistent bias reductions** relative to strong spatial and deep-learning baselines.

Broader Impact:

- Enables **more accurate causal estimates** that can inform policy decisions in applications with both interference and unobserved spatial confounding such as **environmental health, climate science, and social sciences**.

github.com/moprescu/Spatial-Deconfounder